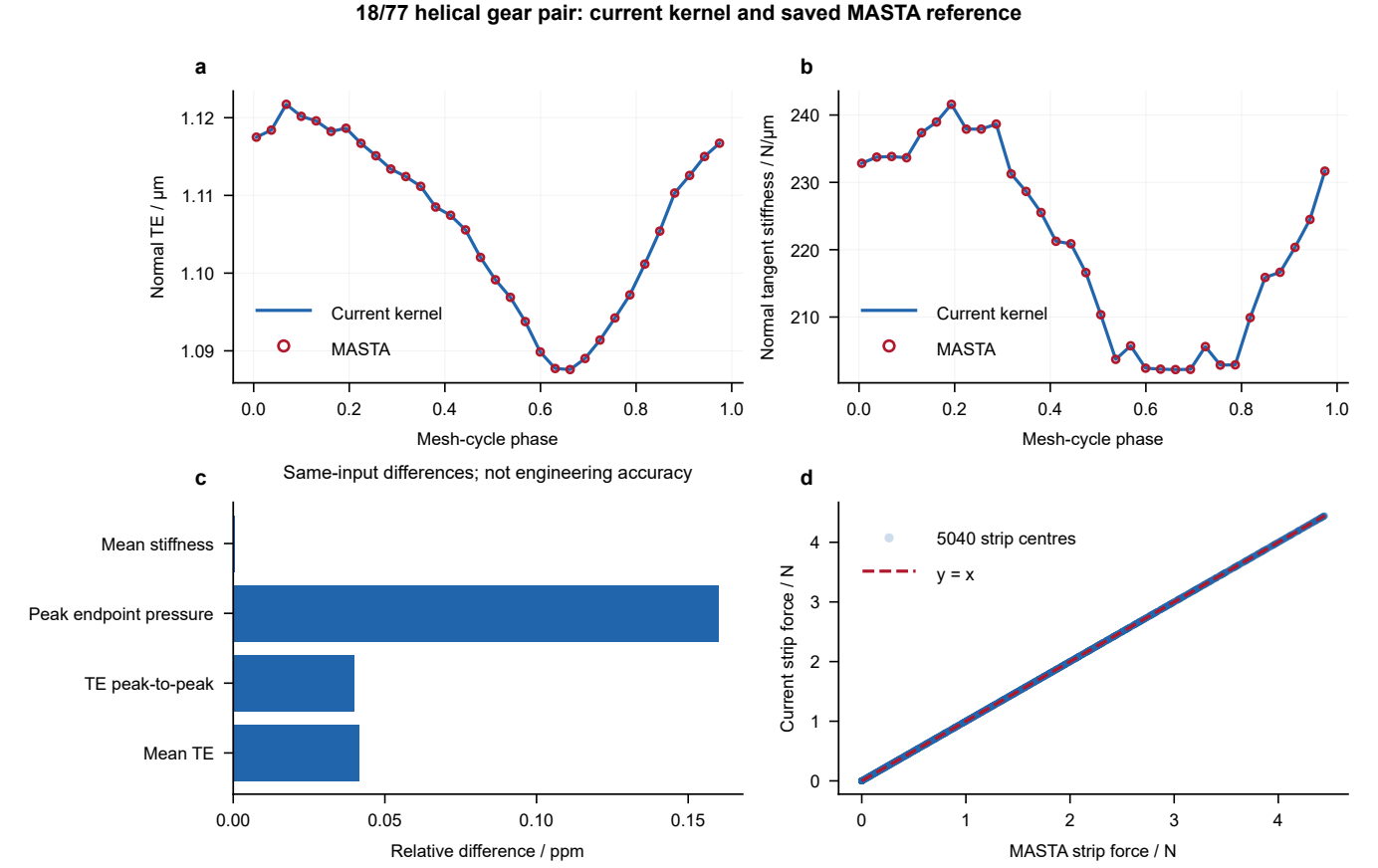



Common inputs and geometry registration; no vendor runtime; no whole-joint or physical validation.